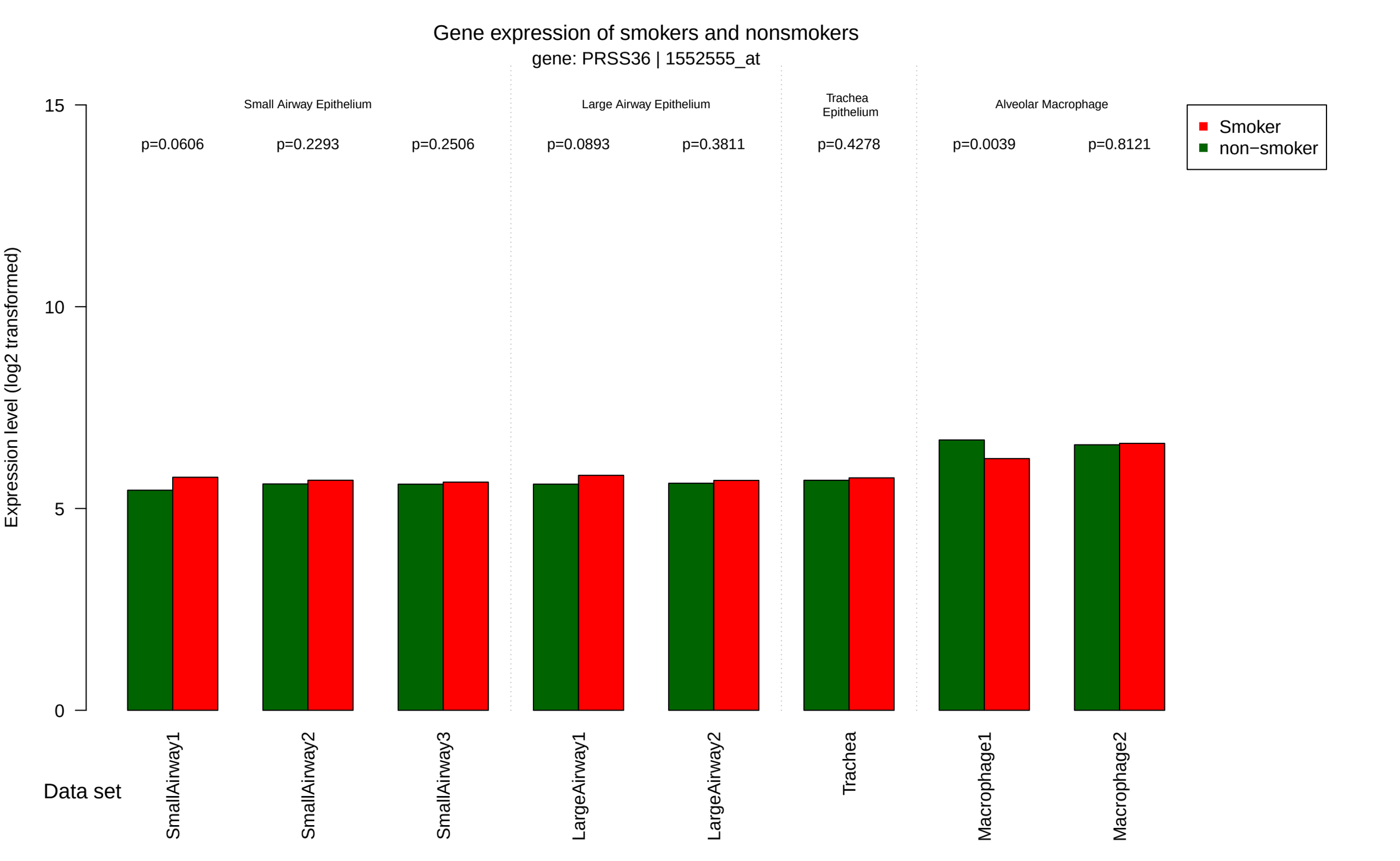
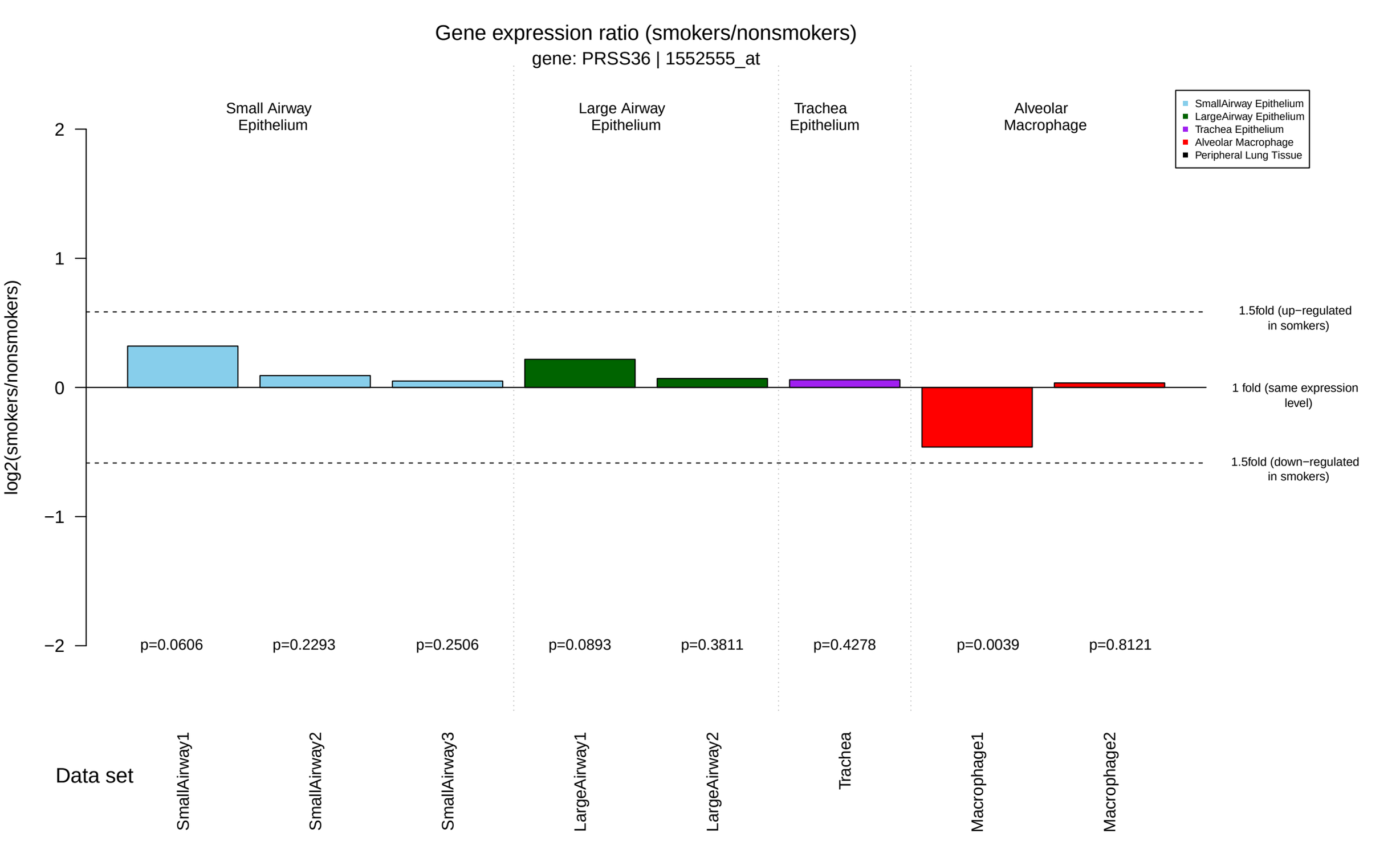
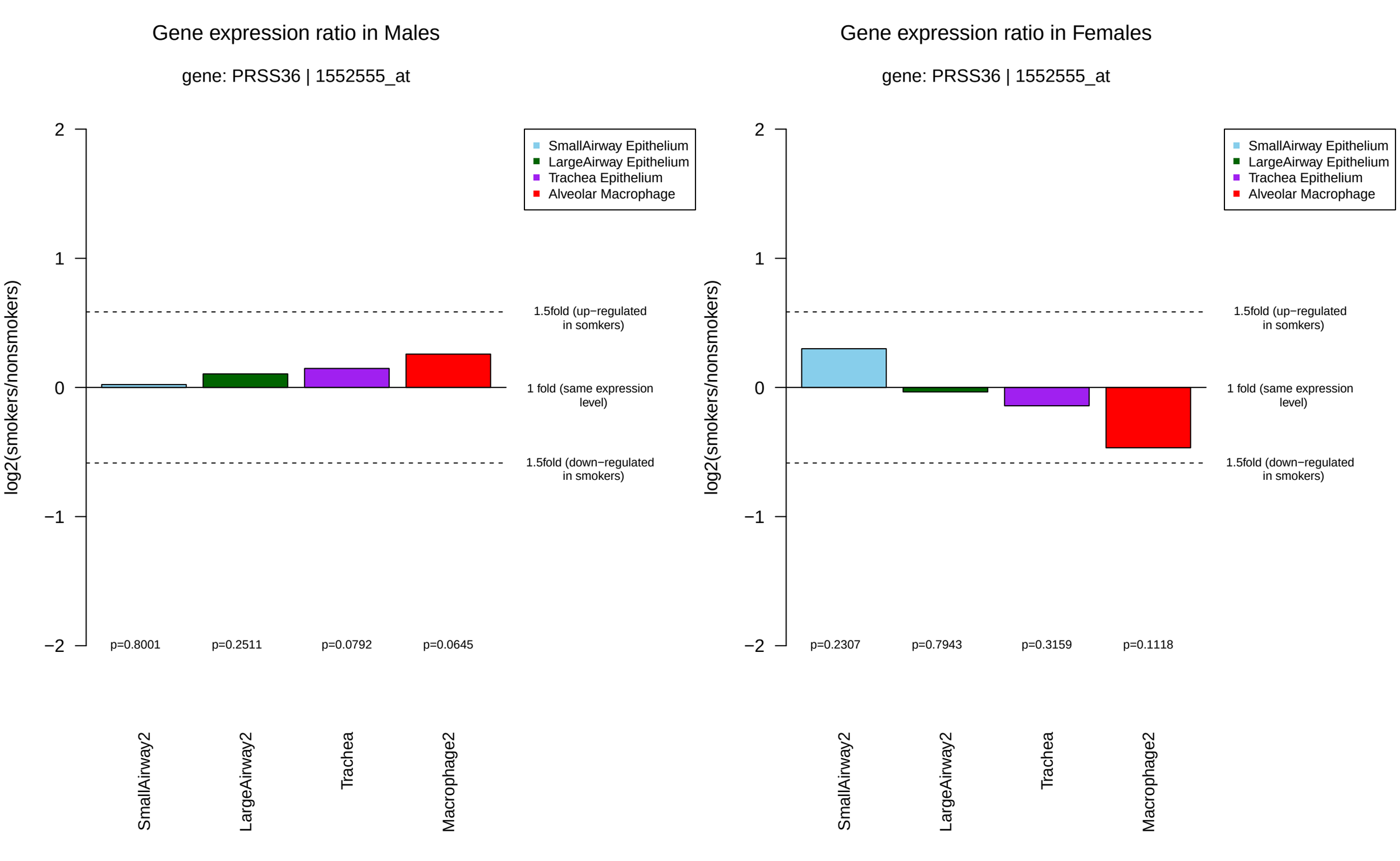
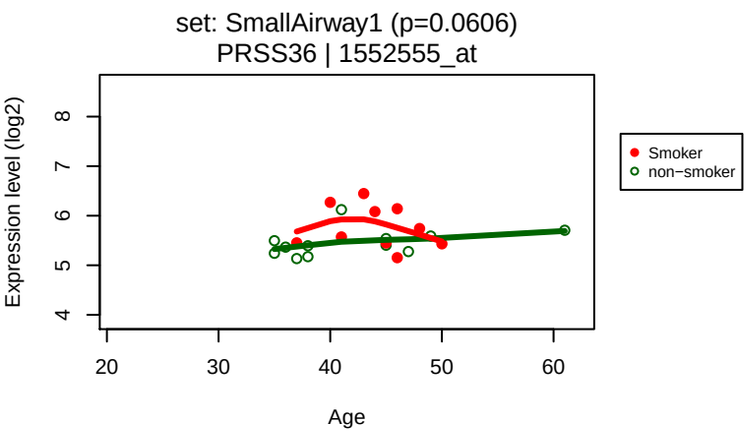
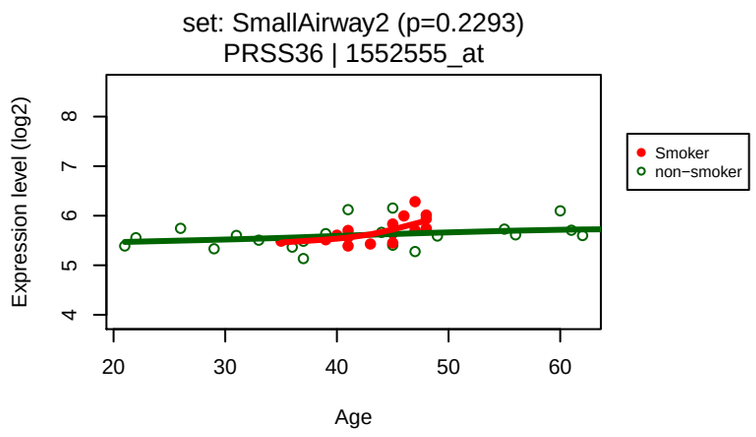
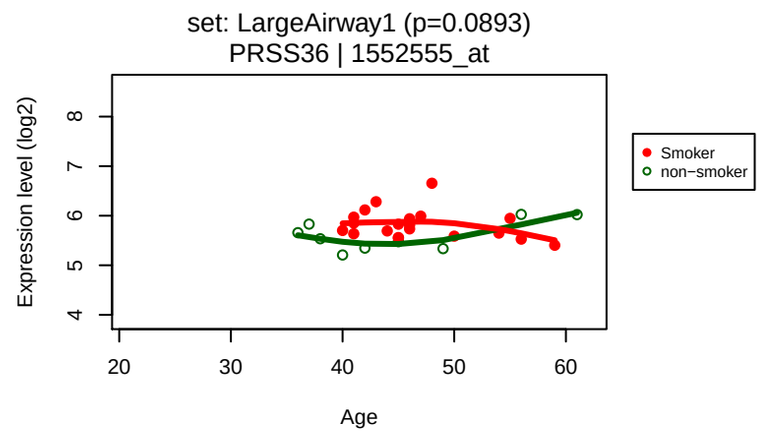

Gene expression of smokers and nonsmokers
gene: PRSS36 | 1552555_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0606
p=0.2293
p=0.2506
p=0.0893
p=0.3811
p=0.4278
p=0.0039
p=0.8121
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

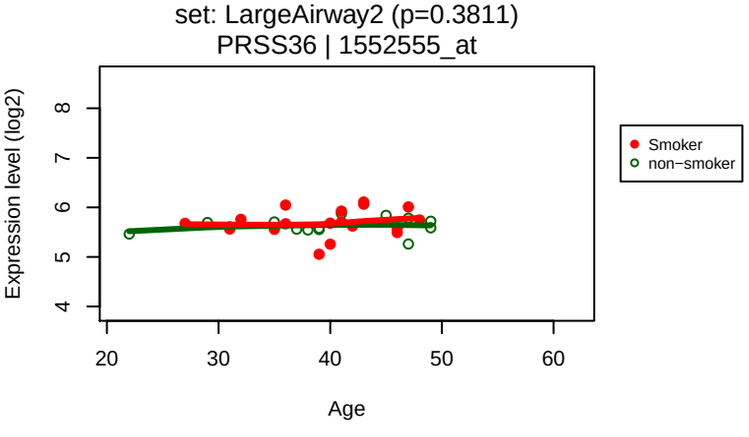

Gene expression ratio (smokers/nonsmokers)
gene: PRSS36 | 1552555_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0606
p=0.2293
p=0.2506
p=0.0893
p=0.3811
p=0.4278
p=0.0039
p=0.8121
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

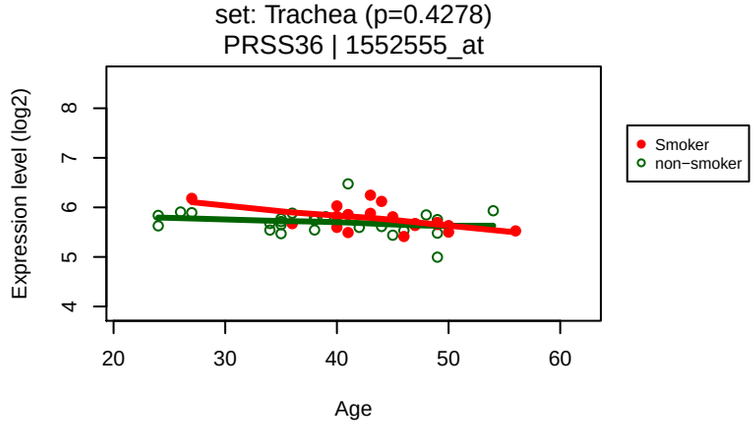

Gene expression ratio in Males
gene: PRSS36 | 1552555_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8001
p=0.2511
p=0.0792
p=0.0645
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS36 | 1552555_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2307
p=0.7943
p=0.3159
p=0.1118
SmallAirway2
LargeAirway2
Trachea
Macrophage2

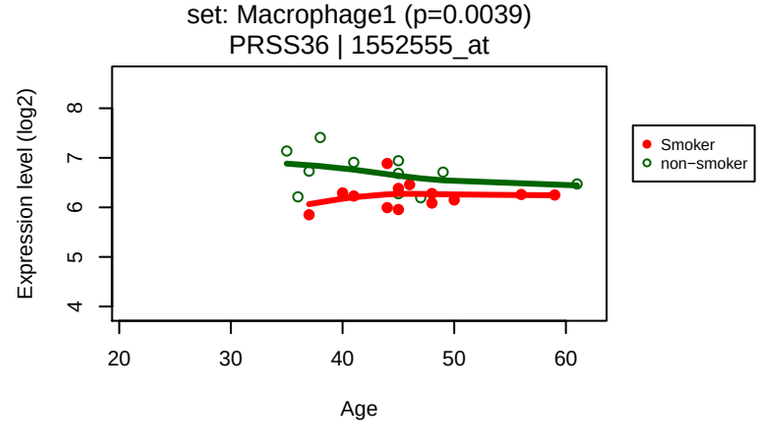

set: SmallAirway1 (p=0.0606)
PRSS36 | 1552555_at
Expression level (log2)
Age
Smoker
non−smoker

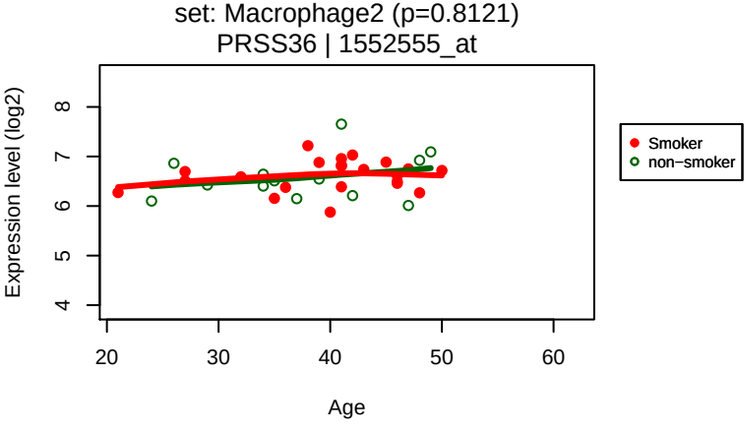

set: SmallAirway2 (p=0.2293)
PRSS36 | 1552555_at
Expression level (log2)
Age
Smoker
non−smoker

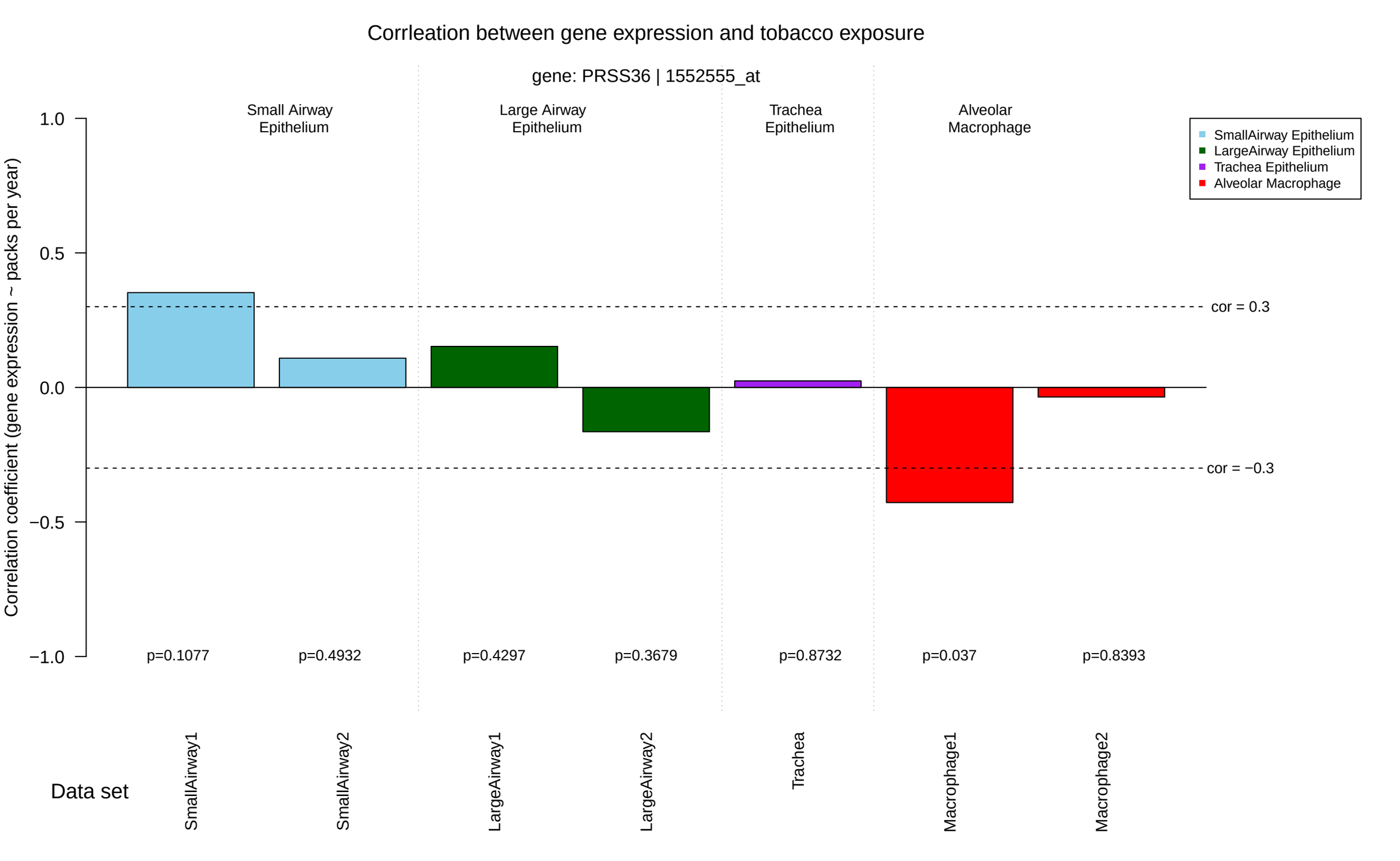

set: LargeAirway1 (p=0.0893)
PRSS36 | 1552555_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3811)
PRSS36 | 1552555_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4278)
PRSS36 | 1552555_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0039)
PRSS36 | 1552555_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8121)
PRSS36 | 1552555_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS36 | 1552555_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1077
p=0.4932
p=0.4297
p=0.3679
p=0.8732
p=0.037
p=0.8393
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2